

Representation Convergence - Project Status

Thesis: independently trained models share representations up to a linear transformation, and that convergence can be deployed. All models frozen; only closed-form linear maps are fitted, except where marked (F).

Models used (all pretrained, frozen unless marked)

Model	Modality / training	Role
GPT-2 (124M)	text, next-token prediction	Exp A subject; E1.1 raw-LM text arm
Pythia-160M	text, next-token (different data/arch lineage)	Exp A counterpart
MobileCLIP-S1	vision, image-text contrastive (phone tier)	Exp B source; F student input
SigLIP 2 base	vision-text contrastive (server tier)	Exp B target; retrieval ceiling; F teacher
DINOv2-base	vision, self-supervised, no language supervision	E-series image arm
bge-m3	text retrieval encoder, contrastive on 1.2B pairs	E1/E2 text arm
MLP heads (trained)	-	B2-bis and F2 nonlinearity controls; F3 unfreezes MobileCLIP last stage (scaffolded)

Experiments and results

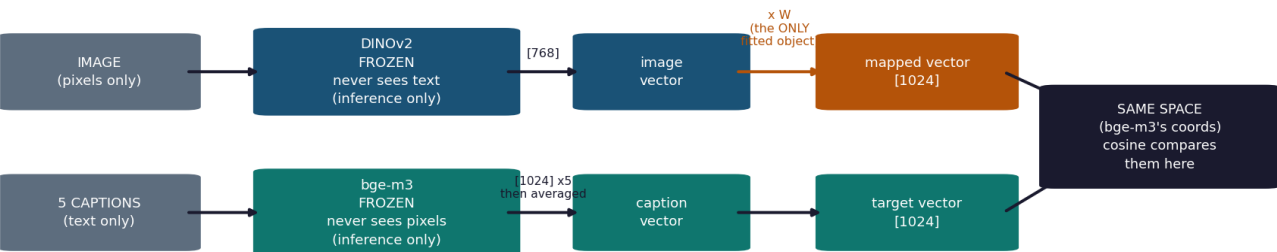
Exp	Question	Setup	Key result	Status
A	Do two LLMs converge?	GPT-2 vs Pythia, 10k WikiText passages, 13 layers, CKA + layer-wise ridge stitching + seeded random-weights control. Data: 7,500 train rows / 768 dims = 9.8	Stitching R2 0.737-0.798 vs random 0.17; CKA 5.3x control; trained and random controls diverge with depth (opposite slopes)	DONE
A2/A3	Is the correspondence structure real?	argmax matrix, lift correction, symmetry, length probes, effective dimensionality	Correspondence elevated but hub-dominated (97% rank-1); lift: 77% within 2 layers; L0 gap = architecture (abs-pos vs RoPE); asymmetry = variance-weighting artifact; eff.dim GPT-2 2.2-10.9 vs Pythia 9.3-92.7 of 768	DONE
B	Are two vision encoders linearly interchangeable?	MobileCLIP -> SigLIP, 4k COCO val2017 pairs, ridge vs Procrustes vs MLP. Data: 3,000 train / 512 = 5.9 (bottom of the 5-10 band)	Ridge R2 0.592 / cosine 0.900; Procrustes fails (0.038) -> anisotropy, confirmed by W's 69x singular-value spread; MLP +0.003 -> relationship is linear	DONE
B3/D	Is it usable?	Retrieval + membership on shared Qdrant index, fp16 export	R@5 = 95.1% of ceiling through a 0.79 MB adapter; raw = chance; cross-tier membership R@1 0.987, gap +0.075 (98.7% positive); fp16 bit-exact	DONE, deployed
B6-B10	Quantization tiers	int8 / int4 adapter + embedding quantization	int8 lossless for this use; documented tier table	DONE
E1	Cross-modal: vision-only vs text-only (PRH's actual claim)	DINOv2 -> bge-m3, 8k COCO train2017, averaged captions, direction- and protocol-matched SigLIP ceiling, pre-registered bands. Data: 7,000 train / 768 = 9.1	R2 0.511, shuffle gap 0.641; R@1 0.358 = 40.9% of matched ceiling (0.875); verdict MEASURABLE BUT WEAK, robust across caption protocols (24.7% / 40.9% both in band); crossing modalities costs ~half of R@1 vs same-modality	DONE
E1.1	Does the text side need similarity training?	Matched pair: replace bge-m3 with GPT-2 mean-pooled; all else identical (8k pairs, 7,000 / 768 = 9.1); target geometry measured before fitting	Predicted split observed: R2 retains 74% (0.377), R@1 retains 27% (0.095, still 95x chance); GPT-2 eff.rank 6.5/768, pair-cos +0.999 -> cosine degenerate (reads 1.000); ridge absorbs anisotropy, cosine cannot	DONE
E1.2	Can label-free geometry rescue GPT-2's space?	6 arms: raw / centered / all-but-top-k / whitened, transforms fitted on train rows only; validated on synthetic control (raw 0.375 -> centered 1.000)	Stake: separates 'correspondence absent' from 'isotropy absent'; caution for cosine-based alignment metrics in the PRH literature	BUILT, not yet run
E2	Interactive demo	image -> nearest captions; text -> nearest images; one W, both directions; negative control included	Working demo over 500 unseen val2017 images	BUILT
E3	Functional stitching: can a consumer USE the map?	SigLIP text tower consumes translated MobileCLIP vectors; 3 arms + data-efficiency curve; internal-splice scaffold	Stitching penalty at R@5 = 0.049 -> near-transparent; 'preserved vs usable' separated	DONE (splice sc scaffolded)

F1-F2	Was the MLP control data-starved?	38k-image teacher corpus, no captions (SigLIP 2, checkpoint teacher-tagged); 38,000 / 512 = 74 rows/dim, 0.029 examples/param for the MLP; relational loss	MLP converges to EXACTLY the linear solution ($0.9052 = 0.9052$, gain -0.0000, plateau by epoch 10) -> bottleneck confirmed twice; data-starvation reading refuted	DONE
F5	Is the refit linear adapter better than the shipped one?	Same ridge refit on 38k rows (74/dim); eval on 5,000 fresh queries, 5 disjoint 1k galleries unseen by both maps; paired bootstrap	All gains REAL: R@1 +0.024 [+0.016,+0.033], R@5 +0.012, R@10 +0.011; same 0.79 MB artifact; 90%-of-ceiling gate approached (89.0-90.6%, dataset-dependent), not cleanly met	DONE
F3-F4	Can the ceiling itself move?	Unfreeze MobileCLIP last stage + head; compare-and-decide with bootstrap ship gate	Gated on F2's verdict (now satisfied); deliberately unrun - trades the adapter's seconds-refit property for weights	BUILT, gated

How the E-series flow works (and what is NOT trained)

Neither model ever sees the other modality. Images enter DINOv2 only; captions enter bge-m3 only. The 'image-caption pair' lives in the row index of two arrays - row i of the image array and row i of the caption array describe the same COCO image - which is bookkeeping for the fit, not an input to either network. **Direction of the map.** W converts DINOv2 vectors [768] into bge-m3's coordinate system [1024] because the space you land in must be the space you search in: image vectors and caption vectors then live in one index and cosine ranks across the modality boundary. One W serves both search directions - for text-to-image, the gallery images are pushed through W once and a typed query meets them natively. Fitting from the narrower side (768 in) also needs fewer rows for the same statistical power. **No training occurred.** Both models are downloaded frozen and run under no_grad; W is a closed-form ridge solution, not gradient descent. After every A/B/E experiment the model weights are bit-identical to the downloads - delete W and nothing remains. The experiments MEASURED a pre-existing relationship; they did not train one in. (The F series is the deliberate exception - the contrast condition, not a lapse: F2 trains a separate MLP head beside the frozen encoders, and F3 - built but deliberately unrun - is the only notebook that would modify a pretrained model's own weights. As of this report every pretrained model in the project remains bit-identical to its download.)

E1 data flow - the two models never see the other modality; only their OUTPUTS meet



the "pairing" (image i <-> its captions) lives in the ROW INDEX of two arrays - it is bookkeeping, not an input to either model

The same flow as arithmetic (toy sizes: image dim 3, text dim 2)

image vec $a = [0.9, 0.1, 0.4]$

target $t = [0.60, 0.80]$

$$W \begin{pmatrix} 3 \times 2 \end{pmatrix} = \begin{bmatrix} 0.5 & 0.6 \\ 0.4 & -0.2 \\ 0.3 & 0.9 \end{bmatrix}$$

fitted by ridge on 7,000 such rows

$$\text{predicted } u = a \times W$$
$$= \begin{bmatrix} 0.9 \times 0.5 + 0.1 \times 0.4 + 0.4 \times 0.3, \\ 0.9 \times 0.6 + 0.1 \times -0.2 + 0.4 \times 0.9 \end{bmatrix}$$
$$= [0.61, 0.88]$$

$$\cos(u, t) = (0.61 \times 0.60 + 0.88 \times 0.80) / (|u| |t|) = 0.999 \rightarrow \text{row } i \text{ found}$$

What exists before and after the experiment

DINOv2 weights	unchanged - bit-identical
bge-m3 weights	unchanged - bit-identical
W (768 x 1024 matrix)	NEW - the only thing created; closed-form ridge, no gradient descent, seconds to fit

delete W and you are back to exactly the two downloaded models - the experiment MEASURED a relationship, it did not train one in

Figure: the E1 flow, the same flow as worked arithmetic (toy dimensions), and the before/after inventory - two unchanged models plus one new matrix.

Caption protocol in the E-series

COCO supplies **five independent human captions per image**, written by five annotators; five is the dataset's structure, not a tuned choice. The decision was whether to use one or all. A single caption carries that annotator's emphasis - one names the activity, another the setting - so the target vector mixes image content with individual phrasing. Averaging the five embeddings cancels the idiosyncrasies and keeps what they agree on, making the target 'what the image is about' rather than 'what one person wrote'. The effect is large: held-out R2 rises 0.278 to 0.511 and R@1 0.160 to 0.358. The same averaging must be applied to the ceiling, because it makes the task easier for any method - SigLIP itself scores 0.648 on single captions and 0.875 on averaged ones. Scoring an averaged system against a single-caption ceiling produced an inflated 55.2 per cent; matched properly it is 40.9. Both protocols are reported, and both fall in the same pre-registered band (24.7 and 40.9 per cent), so the verdict does not depend on the choice; note that the standard COCO benchmark convention is one caption per query, so averaging is a deliberate, disclosed deviation.

What has been established

1. Convergence is real and linear. Two LLMs and two vision encoders, trained independently, are related by a single linear map (held-out R2 0.74 / 0.59); an MLP control adds nothing at 3k samples (+0.003) and converges to exactly the linear solution at 38k (-0.0000). **2. It is not rigid.** Rotation-only alignment fails (0.038); the deviation is anisotropy, measured three independent ways (69x singular-value spread; 58x effective-rank gap; Procrustes-cosine 0.775 with R2 0.038). **3. It is usable.** One 0.79 MB matrix gives two encoders one index at 95% of ceiling, plus an unplanned capability (cross-tier membership 0.987) predicted from the theory and then confirmed. **4. It extends across modalities, weakly.** A vision model with no language supervision maps onto a text encoder at 358x chance - 41% of a jointly trained ceiling - with the verdict robust to protocol choice. **5. Metric validity depends on the space's geometry.** In SigLIP's space cosine was the honest metric and R2 understated; in GPT-2's near-1D space cosine reads 1.000 while retrieval runs at 9.5%. No similarity number is interpretable before the anisotropy battery is run - a caution that applies to the PRH literature's own metrics.

Methodological failures caught and converted

Run-1 sample starvation (5.9 rows/dim) -> the ≥ 5 rule and the power gate; synthetic hot-diagonal figure -> replaced with the real hub-dominated matrix and an honesty note; mismatched E1 ceiling (borrowed direction and protocol) -> measured matched ceiling, verdict corrected 55.2% -> 40.9%; wrong teacher checkpoint in F1 (SigLIP 1 vs 2) -> caught by a transfer check against existing artifacts, now automated; premature 'gate passed' claim -> refuted by expanded paired evaluation (F5). Each failure produced a reusable guard now built into the notebooks.

Deliverables

Final report (25 pp, appendices C-D) - Concepts & Corrections - Project Atlas - Vector Similarity Handbook - defense deck + cheat sheets - 25 runnable Colab notebooks (A1-F5) including V1, a reusable structure/semantics diagnostic battery for any [n, d] representation - deployed artifact: 0.79 MB fp16 adapter, shared Qdrant index, 39 ms end-to-end demo path.

Open items

E1.2 isotropy-rescue run (embeddings ready, minutes); E1 scale sweep (dinov2-small/base/large with capacity controls) - would turn the two-point PRH sample into a slope; F3/F4 if moving the ceiling is ever worth trading the seconds-refit property; E3 internal-splice scaffold.